Katale AI

JUDGE'S GUIDE

Run a small shop from WhatsApp.
The AI prepares. Code handles the money.

Katale AI turns WhatsApp text and voice notes into priced orders, PayPal payment links, stock updates and business insight. The AI only proposes: the backend calculates every price, and payment requests need a typed one-time code from the merchant.

Working demo	kataleai.space
Source code	github.com/yigagilbert/WhatsApp-Voice-Commerce-Agent
Sample shop	kataleai.space/shop/kampala-crafts
Demo video	youtu.be/yVa0QfD9Ixc
Payments	PayPal Sandbox only. No real money moves.

Test it

Steps 1–6 take about 2 minutes in a browser. Sign in at kataleai.space with the sample login credentials in the testing instructions section of the project details on Devpost. It is a shared Owner test account, so please don't unlink WhatsApp or change the API keys. The **Agent console** runs the same AI pipeline as WhatsApp.

1. **Agent console:** send `Which products are running low?` The answer comes from live stock records.
2. **Agent console:** send `Create an order for Sarah for 2 handmade baskets`. The backend prices a draft, and nothing reaches PayPal until an owner types the one-time code.
3. **Products:** click a stock count, change it and press Enter. This is AG Grid inline editing with server validation.
4. **Overview:** click the `How is this week going?` chip, then type `Build me a chart of net revenue by channel`. AG Studio adds the chart to an unsaved AI dashboard tab.
5. **Overview, Products & inventory tab:** right-click a bar in Product revenue and choose **See the records behind this**.
6. **Orders:** open an order, then **View the full trace** to follow message, AI decision, approval and PayPal.
7. **WhatsApp as a customer (no setup, do this before step 8):** open kataleai.space/chat and send `hi`. Pick a product, ask for a better price, then check out. Pay with the Sandbox test buyer in the Devpost testing instructions.
8. **WhatsApp as the merchant:** open **Admins**, add your number in international format (for example `+256700000000`), choose **Owner** and press **Send test**. That number now gets the merchant assistant. Check stock, add a product, ask for a chart, create an order, approve it with the typed code and pay the link. An English voice note works too. Remove your number from Admins when you finish.

Why WhatsApp

Small sellers in Africa already sell in chats, so Katale asks for no new app and no new habit. In a GeoPoll survey (July 2025; 3,945 mobile users aged 18–50 in four African countries), 90% used WhatsApp and 64% used it for business. The sample skews younger and more affluent. The shop also has a public page, with no login, whose link can go in a social media bio. Its cart continues in WhatsApp.

What the video shows

A customer browses, checks out and pays `ORD-K6RQKZ` ($30) in PayPal Sandbox. A separate order shows AI bargaining. The merchant then creates an order, approves it with a typed code, and runs stock, charts and AG Studio. Phone footage is sped up in places, and real customer details are blurred.

How a payment stays safe

1. AI prepares
Understands text and voice, then picks a typed tool.

2. Backend verifies
Customer, product, price, stock and totals from the database.

3. Merchant approves
A one-time code bound to that one order. A spoken "confirm" is refused.

4. PayPal executes
Checkout, capture and signature-verified webhooks.

How bargaining stays safe

Customers can haggle on WhatsApp, as they would at the market. For each product, the merchant can set a private lowest price. The AI chooses how to say an offer, never the number: the backend calculates each one, and the AI never sees the minimum. A price guard checks every amount in an AI reply before it is sent, and the agreed price is what goes to PayPal.

How we use PayPal

Orders v2	Creates the checkout link for each approved order and captures the payment.
Webhooks	Verified by signature and idempotent. They mark orders paid, record refunds and raise dispute alerts.
Order tracking	Sends carrier and tracking numbers when the merchant ships. Built, not shown in the video.
Disputes v1	Lists cases, then submits the merchant's response after a typed code. Built, not shown in the video.
Reconciliation	Checks open payment links against PayPal.

How we use AG Grid and AG Studio

AG Grid	Orders, products, payments, customers and disputes grids, with filters, sorting, CSV export and inline editing of price and stock.
AG Studio	The Overview dashboard: report pages, filters and right-click drill-through.
AG Studio AI	The assistant answers from deterministic lookups and adds charts from a checked list of metrics. Nothing is kept until the merchant saves it.

Stack

Next.js and Tailwind dashboard; Fastify API; PostgreSQL with Prisma; whatsapp-web.js worker; an OpenAI-compatible LLM (`gpt-4.1-mini`) with typed tools; OpenAI Whisper for voice notes; Docker deployment.

Known limits

- **Where it works:** any WhatsApp number, but receiving money needs a PayPal business account that can receive payments in the seller's country. PayPal doesn't support UGX, KES, TZS or RWF, so prices are in USD or another PayPal currency.
- **WhatsApp connection:** for now we use whatsapp-web.js, an unofficial client that WhatsApp's terms don't allow and that can get a number blocked. We plan to move to the official WhatsApp Business API soon after company registration.
- **Voice:** English is the supported language. The voice note in the video was a test take with a synthetic voice.
- **Accounts:** one merchant per installation, with no self-serve multi-merchant sign-up yet.